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PHINMA EDUCATION

BLOOD PROCESSING (Pretransfusion Reaction)

BLOOD BANKING

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DONOR BLEEDING

45 (10 - 20

- Standard needle used: 16 gauge
- Needle insertion:
  - ➢ For blood collection, most blood centers use an iodine compound such as PVP-iodine or polymer iodine complex. Using a tourniquet or blood pressure cuff, the venipuncture is identify and the area is scrubbed at least 4 cm in all directions from the site for a minimum of seconds
  - ➢ For donor blood collection, Scrub the site (2x2 inches) for 30 seconds using an aqueous iodophor scrub solution (0.7%).
- Donations of whole blood where the bleed time exceeded 15 minutes are not suitable for the production of plasma components, platelet products, and cryoprecipitate for direct clinical use.

DONATION PROCESS

1. Donor registration
2. Interview and physical exam
3. Donor selection and blood collection

I. BASIC QUALIFICATIONS OF THE POTENTIAL BLOOD DONOR

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1. **GENERAL APPEARANCE**
   - Donor center Representative should observe the prospective donor for presence of excessive anxiety, drug or alcohol influence, or nervousness.
1. **DOB/AGE**
   - At least **18-65 years**
     - <18 y.o. - need parent's consent
     - **>65 y.o. - need physician's consent**
1. **WEIGHT**
   - Atleast 110lb (50 kg)
   - Maximum of 10.5 mL of blood/kg of donor weight
   - Donors who less than 50kg are NOT restricted from donating
     - ❑ First formula

A. VOLUME OF BLOOD TO BE DRAWN (mL)

$$\frac{\text{Donor's weight x 450 mL}}{\text{Ideal weight}} = A$$

B. VOLUME OF ANTICOAGULANT NEEDED (mL)

$$\frac{\text{Answer in A x 14}}{100} = B$$

C. VOLUME OF ANTICOAGULANT TO BE REMOVE FROM THE BLOOD BAG (mL)

$$\text{63 mL – Answer in B} = C$$

3. WEIGHT

- From Harmening:
  - **Volume of blood to collect** = (donor' s weight in kg/50) x 450 mL
  - **Volume of anticoagulant needed** = volume to collect / 450 x 63 mL
  - **Volume of Anticoagulant to be removed from blood bag** = 63 mL - above calculated volume = amount of solution to be removed.

4. TEMPERATURE

- Must be less than or equal **99.5° F (37.5°C) orally**

5. PULSE

- Donor' s pulse should be between 50 and 100 beats per minute

6. BLOOD PRESSURE

- Potential donor' s systolic blood pressure should be **LESS THAN or EQUAL to 180 mmHg** and the diastolic **LESS THAN or EQUAL to 100 mmHg**

7. HEMOGLOBIN & HEMATOCRIT

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- ***Allogeneic donation***
  - Women:
    - Hemoglobin: **≥ 12.5 g/dL**
    - Hematocrit: **≥ 38%**
  - Men:
    - Hemoglobin: **≥ 13.0 g/dL**
    - Hematocrit: **≥ 39%**
- ***Autologous donation***
  - Hemoglobin: **≥ 11.0 g/dL**
  - Hematocrit: **≥ 33%**
    - **>1.053** copper sulfate specific gravity

CuSO4 Method

- ***Principle:*** A drop of whole blood when dropped in a solution of copper sulfate solution will maintain its density for approximately **15 secs**.
- A 30 ml container of copper sulfate can perform **25 tests**.
- The distance between the drop of blood and copper sulfate solution is 1 cm.

8. SKIN LESIONS

- Prior to donation, the donor' s arms should be inspected for skin lesions. Evidence of skin lesions (e.g., Multiple puncture marks) is cause for indefinite deferral.

Additional information:

> Name of the patient for whom the blood is intended (directed donation)
> Race of the donor for matching specific phenotypes
> Cytomegalovirus (CMV) status (some patient groups, such as neonates, require CMV-negative blood in certain circumstances).

COLLECTION OF BLOOD

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1. SKIN PREPARATION	• Aseptic method
2. VOLUME OF BLOOD ROUTINELY COLLECTED	• 450 mL +/- 10% or 500 mL +/- 10%, depending on the collection bag • Unit containing a volume 405 to 550 mL should be weigh between 429 to 583 g, plus the weight of the container and anticoagulant
3. MAXIMUM VOLUME	• 10.5 mL of blood per kg of donor' s weight, including samples for testing
4. LOW-VOLUMES COLLECTIONS	• 300 to 404 mL in 450 mL or 333 to 449 mL in 500 mL bag
5. VOLUMES OF ANTICOAGULANT	• 63 mL anticoagulant for 450 mL collection, 70 mL for 500 mL collection
6. STORAGE TEMPERATURE OF UNIT BETWEEN COLLECTION AND PROCESSING	• All whole blood units should be stored at 1° - 6° C • For platelet production should be stored at 20° - 24 ° C, until platelets have been removed

CURRENT DONOR SCREENING TESTS FOR INFECTIOUS DISEASES

TEST	DATE TEST REQUIRED
Syphilis	1950s
Hepatitis B surface antigen (HBsAg)	1971
Hepatitis B core antibody (anti-HBc)	1986
Hepatitis C virus antibody (anti-HCV)	1990
Human immunodeficiency virus antibodies (anti-HIV-1/2)	1992
Human T-cell lymphotropic virus antibody (anti-HTLV-I/II)	1997
Human immunodeficiency virus (HIV-1) NAT	1999
Hepatitis C virus (HCV) NAT	1999
West Nile virus NAT	2004
Trypanosoma cruzi antibody (anti-*T. cruzi*)	2007
Hepatitis B virus (HBV) NAT	2009
Babesia microti antibody and NAT (recommended)	2012
Zika virus NAT	2016

DONATION INTERVAL

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DOH 12 weeks from the last donation; shorter interval may be allowed provided
8 weeks after whole blood donation
At least 2 days after plasma, platelet, or leukocyte apheresis
4 weeks after infrequent apheresis
16 weeks after two-unit red cell collection (DOUBLE RBC PHERESIS)

TYPES OF DEFERRAL

TEMPORARY	• Prospective donor is unable to donate blood for a limited period of time.
INDEFINITE	• Prospective donor is unable to donate blood for someone else for an unspecified period of time due to current regulatory requirements. • would not be able to donate blood until the current requirement changes.
PERMANENT	• Prospective donor will never be eligible to donate blood for someone else. • may be eligible to donate autologous blood. • Some permanent deferrals may result from the testing performed on a previous

DONOR DEFERRAL

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24 HOURS	• Donors drunk with alcoholic beverage
48 HOURS	• Whole blood donation deferred after hemapheresis • Ingestion of irreversible platelet function inhibitors (e.g., Aspirin or Piroxicam) • Donor taking Feldene
72 HOURS	• Dental extraction, uncomplicated (DOH)
2 WEEKS	• Donor taking Plavix and Ticlid • After vaccination with live attenuated viral or bacterial vaccines such as: Oral typhoid, Oral Polio (Sabin), Measles (rubeola), Mumps, Yellow fever, Influenza
14 to 21 DAYS	• Small pox vaccination until the scab has fallen off
1 MONTH or 4 WEEKS	• After vaccination of live attenuated vaccine for German measles (Rubella), Varicella zoster (Chicken pox) and MMR vaccination (Harmening) • After cessation of drugs: Isotretinoin (Accutane) used for acne treatment, Finasteride (Proscar, Propecia) used for treatment of benign prostatic hyperplasia or alopecia
6 WEEKS	• Following delivery of baby (AABB)
8 WEEKS / 2 MONTHS	• Interval between whole blood donation • MMR vaccination (DOH) • When RBC's are collected by pheresis

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6 MONTHS	• Donor taking Avodart • Individuals with a history of dengue and chikungunya virus
12 MONTH / 1 YEAR	• After HBIG administration • After therapeutic rabies vaccination • Rape victims • Healthcare workers with percutaneous or mucous membrane exposure to blood or body fluids • Patients who have received blood, blood components, blood derivatives, or human tissues known to be a source of blood-borne pathogens • Close contact with a patient with viral hepatitis • Application of a tattoo • Sexual contact with a prostitute or other persons on a high-risk group for AIDS or HIV positive individual, Hepatitis, with hemophiliacs, or with IV drugs users • Female who had sexual contact or men who ever had sex with another men • After exposure to or bite by a rabid animal • Incarceration (imprisonment) in a correctional institution for more than 72 consecutive hours • Intranasal use of cocaine • Residents of nonendemic areas who travel to endemic areas of malaria • History of syphilis or gonorrhea, treatment, or a reactive screening test for syphilis in the absence of a negative confirmatory test • Needle stick injury • Ear and body piercing • Male sexual contact with another male • History of Leishmaniasis

12 MONTH / 1 YEAR	• 12-month deferral would apply if the woman received a transfusion during her pregnancy
3 YEARS	• Use of the drug Acitretin (Soriatane)
PERMANENT	• Those who have taken teratogenic drug Etretinate (Tegison) for the treatment of psoriasis • Family history of CJD • Treatment with pituitary growth hormone of human origin because of the possibility of transmitting CJD (CA: Prions) PRION DISEASES: 1. Scrapie-infects brain of sheep and goats and they scrape theirs against a pole 2. Kuru-cannibalism: from eating infected monkey brains; seen in New Guinea 3. CJD 4. Bovin Spongiform Encephalopathy (Mad Cow Disease) • Parenteral drug abuse • History of hemophilia A or B, von Willebrand disease, or severe thrombocytopenia • Viral hepatitis after 11th birthday • Confirmed positive HBsAg, HCV, HTLV, or HIV • Repeatedly reactive anti-HBc on >1 occasion • Past or present clinical or laboratory evidence of infection with HIV, HTLV, OR HCV

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PERMANENT	• History of babesiosis or Chaga's disease • Received dura mater transplants • Xenotransplantation • High risk history for AIDS: ❖ men who have had sex w/. another man anytime since 1977 ❖ hemophiliacs ❖ Intravenous drug abuser (past or present) ❖ Person who have engage in sex for money or drugs anytime since 1977

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INDEFINITE	• Born or lived in any of the African countries on the FDA list of countries with HIV-1 group O risk since 1977 ❖ who were born in Cameroon or surrounding countries ❖ residents of these countries since 1977 ❖ travelers to those countries who received blood transfusions and their sexual contacts • Sexual contact with anyone who was born in or lived in Africa • History of cancer, leukemia, or lymphoma
PERMANENT/INDEFINITE	• Chemotherapeutic agents administered for malignancy • Chronic cardiac, pulmonary, liver, or renal diseases • Donor implicated in a case of recipient's post-transfusion hepatitis, HIV, or HTLV infection • Those who have received an injectable product from a British bovine source (i.e, British Bovine Insulin)

NO DEFERRAL	• Immunization with recombinant vaccines, toxoids or dead organism if asymptomatic including primary prophylaxis against: ❖ Anthrax, Cholera, Diphtheria, HAV, HBV, Influenza, Lyme disease, Typhoid infection, Paratyphoid, Pertussis, Plague, Pneumococcus, Polio (Salk), Rabies, Rocky mountain spotted fever, Tetanus • First-trimester or second-trimester abortion or miscarriage

ADDITIONAL FROM DOH	
No deferral, may donate anytime	• DM: controlled by diet and oral medication
Permanent	• DM: uncontrolled and on insulin
72 Hours / 3 Days	• Dental extraction, uncomplicated
1 Month	• Dental extraction, with abscess
1 Month	• Dengue Fever
8 Weeks	• MMR vaccination
12 Months / 1 Year	• Dental surgery
24 Months / 2 Years	• Tuberculosis

12 weeks / 3 months	• Interval between whole blood donation

MALARIA

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FROM AABB	
3 YEARS	• History of malaria
3 YEARS	• Immigrants, refugees, citizens, or person who have resided in the endemic area for at least 5 consecutive years
1 YEAR	• Travelers to area the CDC considers to be endemic for malaria
FROM DOH	
3 YEARS	• Malaria infection
6 MONTHS	• Travel to area endemic with or without prophylaxis (stay < 6 months)
12 MONTHS	• Former resident of malaria endemic area, or stayed in malaria endemic area for 6 months or more
DOH MEMO NO. 2020-0124 INTERIM GUIDELINES ON ENSURING BLOOD SAFETY AND ADEQUACY FOR COVID-19 REPONSE	
28 DAYS	• History of travel to any countries with Covid-19 local transmission • Close contact with Person Under Investigation (PUI), Person Under Monitoring (PUM) or with confirmed Covid-19 case
14 DAYS	• Respiratory Tract Infection symptoms (cough and clods)

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DOH CIRCULAR NO. 2021-0102A UPDATING OF TIMING OF BLOOD DONATION WHEN DONORS RECEIVE COVID-19 VACCINE	
14 DAYS AFTER VACCINATION	• Unknown vaccine type
Accept anytime if asymptomatic	• Moderna, Pfizer BioNTech, Novavax, AstraZeneca, Janssen, Coronavac, Covaxin

RETENTION OF DONOR RECORDS

DONOR RECORD		RETENTION TIME (YEARS)
• Medical director approval for donation interval		5
• Donor ABO/Rh • Donor antibody screen • Informed consent for donation • Physical examination • Medical history information • Identification number of donor unit	• Viral marker testing results • Quarantine of donor unit • Repeat testing of donor blood • ID of donor-processing tech • Plt count for frequent platelet apheresis • Sedimenting agent of leukapheresis	10
• Notification of abnormal results		10/indefinite

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Donor card and consent forms	• Need to be signed by the qualified donor prior to phlebotomy
7–10 minutes	• Duration of bleeding a donor
4 hours (60 drops/min)	• Duration of blood transfusion to prevent bacterial contamination
At least 7 days	• Post transfusion blood specimens should be kept stored at 1 -6'C
Hepatitis C	• Cause of majority of hepatitis transmitted through blood transfusion
Free of skin lesions	• The area of venipuncture that is selected
Stop venipuncture at once	• When a donor shows adverse rxn or symptoms during phleb
ATP	• Component most important for the viability of RBC
Mannitol	• Component in additive solutions used for red cell preparation serves as stabilizing agaent
Vitamins (A, D, E, K) (Factors IX, X, VII, II)	• Fat soluble vitamins
Bleeding diathesis, citrate toxicity and circulatory overload	• Are problems that may arise in massive transfusion

Rebreathe air from a paper bag	• Indication given to blood donor experiencing rapid breathing and involuntary twitching on his fingers after starting phlebotomy
Reduce flow rate	• Employed on donor experiencing perioral paresthesia during an apheresis procedure
60-150 minutes	• Amount of time for complete apheresis

PREVENTION AND MANAGEMENT OF COMMON DONOR REACTIONS

COMMON DONOR REACTIONS	COMMON ATTRIBUTED TO	PREVENTION/MANAGEMENT
1. Lightheadedness 2. Weakness 3. Tingling sensation 4. Palpitations	Anxiety Hypoglycemia	Reassuring conversation Try any of the following: 1. Elevate donor's feet to 45' angle (few minutes) then lower to 20' angle to increase venous return 2. Apply cold, wet towels to neck and forehead 3. Have donor breathe into paper bag 4. Provide juice even before donation Last option: Discontinue donation

Fainting	Anxiety and/or Hypoglycemia	• Discontinue donation • If necessary, administer glucose solution • Provide juice even before donation • Position donor in a place protected from a possible fall
Convulsion	Anxiety or underlying disease	• Discontinue donation • Elevate feet and maintain airway • Restrain gently to prevent injury • Reassure after recovering consciousness • Inform about possible involuntary loss of control of urine or stool during convulsion
Cardiopulmonary emergency	Underlying heart disease	• Ventilation • Cardiopulmonary resuscitation • Transfer donor to Emergency Medical Facility
Hematoma	Very agile veins Unskilled phlebotomist or, Uncooperative donor	• Discontinue if large • Apply pressure to site for atleast 5 minutes • Apply cold packs and reassure donor

Jet-like pulsating bleeding with bright red blood	Inadvertent puncture of artery when deep vein punctures are attempted	• Discontinue immediately , Apply dressing on site • Apply firm pressure over puncture site of atleast 10 minutes • Follow-up donor for additional care of necessary
Shooting pain followed by numbness and tingling in the	Inadvertent puncture of median nerve or cutaneous branches	• Reassure and apply support to the arm

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REQUIREMENTS FOR ACCEPTING UNUSED BLOOD

1. The unit has not been entered or punctured
2. The unit has not been warmed
3. The unit has been returned to the blood bank within 30 minutes of time it was issued (its temp must not exceed 10' C
4. The unit has not been stored in an unmonitored refrigerator

PRETRANSFUSION TESTING

- Use of serologic principles and test to ensure compatibility and prevent an immune -mediated hemolytic transfusion reaction
- Intravenous line (IV) contamination is unacceptable
- Hemolyzed sample should be avoided
- If a patient has been transfused or pregnant with the past 3 months or if the patient's history is unknown, a specimen just be collected every 3 days for antibody screening and compatibility testing. The date of collection is considered day 0, and the specimen expires at midnight on Day 3.
- Pretransfusion specimens and donor unit segment: stored at 2° to 8°C and must be retained in the event that additional testing is warranted.
- T e patient sample and a segment from the donor unit must be retained post -transfusion for at least 7 days

SPECIMEN REQUIREMENTS

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1. **Anticoagulated specimens**
   - Preferred due to ease of handling
   - Ideal for preparing a uniform cell suspension for testing
   - Can inactivate complement-binding antibodies due to calcium chelation disrupting the complement cascade
1. **Clotted RBC's**
   - May require additional washing steps to minimize interference in test interpretations
   - Serum may contain small fibrin clots that can be difficult to distinguish from true agglutination

PRETRANSFUSION TESTING MUST KNOWS

- Pretransfusion testing begins and ends with the proper identification and collection of the patient sample
- A major cause of transfusion-associated fatalities is CLERICAL ERROR resulting in incorrect ABO groupings and transfusion of ABO incompatible blood. Clerical error is the greatest threat to safe transfusion therapy. Most common cause of error is MISIDENTIFICATION of the recipient

PRETRANSFUSION TESTING MUST KNOWS

- Serum and plasma may be used for pretransfusion testing. Disadvantages to using plasma include the possible formation of small fibrin clots, which may be difficult to distinguish from true agglutination
- A 10ml of blood is usually sufficient for all testing procedures if there are no known serologic problems
- A 2 to 5% saline suspension of RBC's is used for most serologic testing procedures
- Both donor and recipient samples must be stored for a minimum of 7 days following transfusion
- In extreme emergencies, when there is no time to obtain and test a pretransfusion sample, group O rh-negative packed cells can be used
- According to AABB standards, a pretransfusion specimen for testing and red cell transfusion is valid for 72 hours. After 72 hours, a new sample must be drawn.

ANTIBODY SCREENING (Detection and Identification

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what is the purpose of Antibody Screening?

- To detect the "Irregular" or "Unexpected" antibodies

TYPES OF UNEXPECTED ANTIBODIES	
Immune alloantibodies	Results of RBC stimulation in the patient
Naturally occurring alloantibodies	Result of exposure to environmental sources, such as pollen, fungus, and bacteria, which have structures similar to some RBC antigens Produced without RBC stimulation
Passively acquired antibodies	Produced in one individual and then transmitted to another individual via plasma-containing blood components or derivatives such as intravenous immunoglobulin (IVIG)
Autoantibodies	Autoantibodies are antibodies directed against antigens expressed on one's own RBCs and generally react with all RBCs tested.
~~Alloantibodies~~	~~Directed at one's own antigens~~
Clinically significant antibody	Results in the shortened survival of RBC's possessing the target antigen Typically IgG antibodies React best at 37°c or in the AHG phase of testing Known to cause Hemolytic Transfusion Reaction and/or HDFN

ANTIBODY SCREENING (Detection and Identification

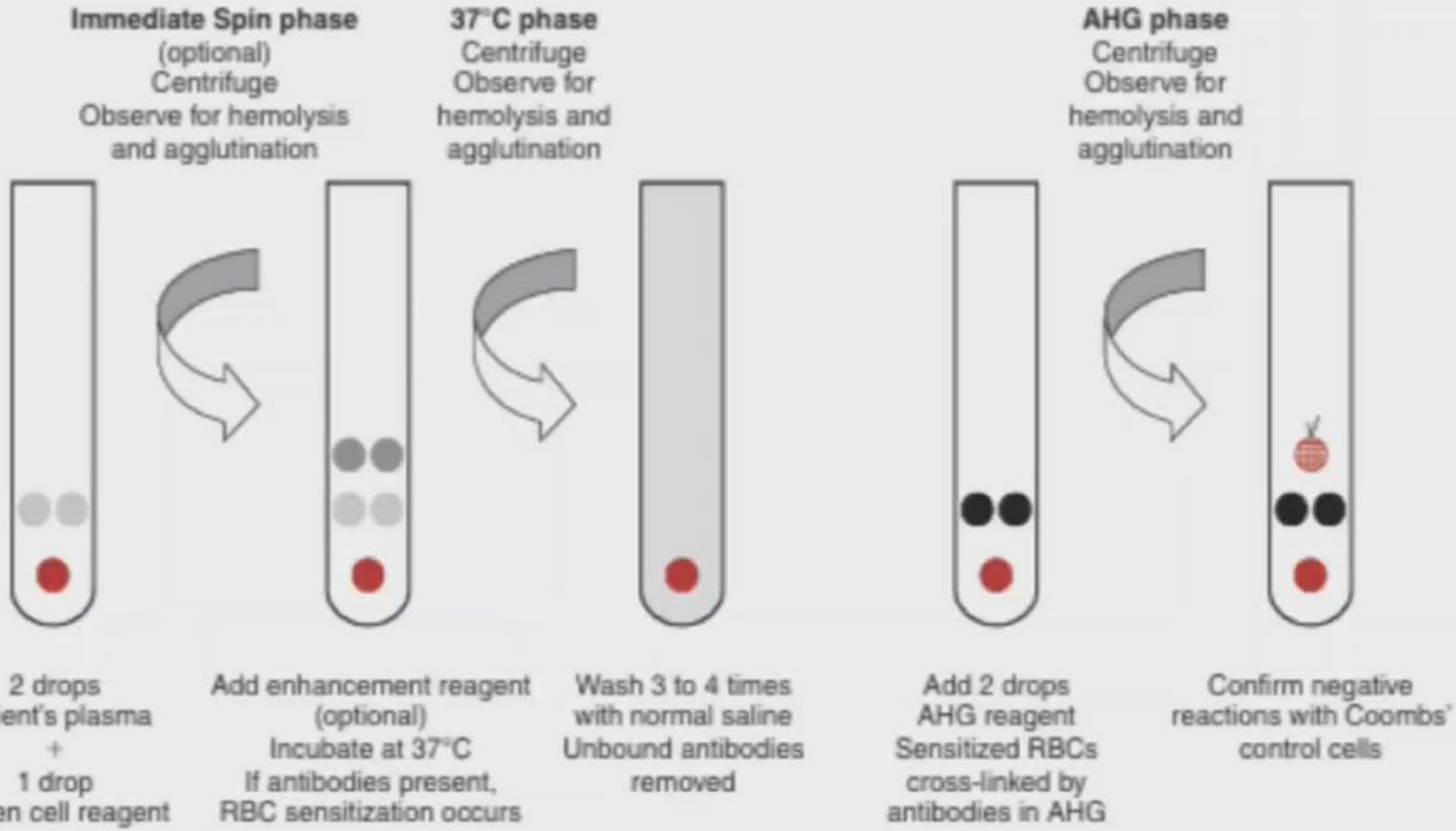
- Relative quantity of an antibody can be determined by testing serial two-fold dilutions of serum against antigen-positive RBC's; the reciprocal of the highest serum dilution showing agglutination is the antibody titer

ANTIBODY SCREENING METHODS

1. TUBE METHOD

REAGENTS USED:	
Screen cells / RBC reagents	Group O individuals who have been typed for the most common and the most significant RBC antigens
Enhancement reagents	LISS, 22% Albumin, and PEG
AHG reagents	Any tube test that is negative after adding the AHG reagent should be controlled by adding Coomb's control cells, which are group O rh-positive RBC's that have been coated with anti-D

- It uses Immediate spin, 37'C incubation phase and AHG


Immediate Spin phase
(optional)
Centrifuge
Observe for hemolysis
and agglutination
37°C phase
Centrifuge
Observe for
hemolysis and
agglutination
AHG phase
Centrifuge
Observe for
hemolysis and
agglutination
2 drops
patient's plasma
+
1 drop
screen cell reagent
Add enhancement reagent
(optional)
Incubate at 37°C
If antibodies present,
RBC sensitization occurs
Wash 3 to 4 times
with normal saline
Unbound antibodies
removed
Add 2 drops
AHG reagent
Sensitized RBCs
cross-linked by
antibodies in AHG
Confirm negative
reactions with Coombs'
control cells

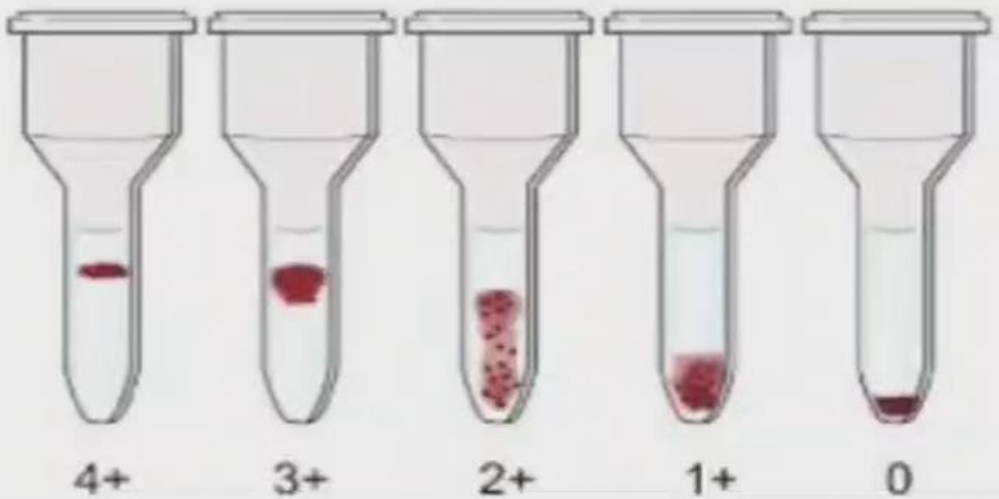
ANTIBODY SCREENING METHODS

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2. GEL TECHNOLOGY AND SOLID PHASE TECHNOLOGY

2.1. GEL TECHNOLOGY

- Invented by Yves Lapierre in 1985 of Lyon France
- Based on the controlled centrifugation of RBC's through DEXTRAN ACRYLAMIDE GEL
- Can do ABO forward and reverse grouping, Rh typing, DAT, Antibod screening, Antibody identification, and Compatibility testing
- Centrifugation time: 10 minutes at 910 rpm
- Incubation time: 15 minutes
- Principle: Hemagglutination





Agglutination Reaction

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4+	Solid band of agglutinated red cells at the TOP of the gel column. Usually no red cells are available in the bottom of microtube
3+	Predominant amount of agglutinated red cells towards the top of the gel column with a few agglutinates staggered below the thicker band. The majority of agglutinates are observed in the top half of the gel column
2+	Red cells agglutinates dispersed throughout the gel column with few agglutinates at the bottom of the microtubes. Agglutinates should be distributed through the upper and lower halves of the gel
1+	Red cells agglutinates predominantly observed in the lower half of the gel column with red cells also in the bottom. These reactions may be weak, with a few agglutinates remaining in the gel area just above the red cell pellet in the bottom of the microtube
NEGATIVE	Red cell forming a well-delineated pellet in the bottom of the microtube. The gel above the red cell pellet is clear and free of agglutinates
MIXED-FIELD	Layer of the red cells agglutinates at the top of the gel column accompanied by a pellet of non agglutinated cells in the bottom of the microtube

2.2. SOLID PHASE TECHNOLOGY

- Based on solid phase red cell adherence
- Can do antibody screening, antibody identification and compatibility testing

2.3. AFFINITY COLUMN TECHNOLOGY

- Principle of affinity adherence if IgG sensitized erythrocytes to an immunologically active matrix
- Can do antibody screening, antibody identification and compatibility testing

IMPORTANT REMINDERS:

- Positive result at any phase of testing indicates the need for antibody identification studies. (GEL TECH METHOD)
- When an unexpected antibody is detected in the antibody screen, an antibody identification panel is performed to determine the specificity of the antibody is present. (TUBE METHOD)

CAUSES OF POSITIVE PRETRANSFUSION TESTS

• Negative Antibody Screen • Incompatible Immediate Spin Crossmatch	○ Donor red blood cells are **ABO incompatible**. ○ Donor red blood cells are **polyagglutinable.** ○ Anti-A1 is the serum of an A2 or A2B individual. ○ Other alloantibodies reactive at room temperature (e.g., anti-M). ○ **Rouleaux formation.** ○ Cold autoantibodies (e.g., anti-I). ○ Passively acquired anti-A or anti-B.
• Negative Antibody Screen • Incompatible Antiglobulin Crossmatch	○ Donor red blood cells have a **positive direct antiglobulin test.** ○ Antibody reacts only with red blood cells having strong expression of a particular antigen (e.g., dosage) or variation in antigen strength (e.g., P1). ○ Antibody to a low-incidence antigen on the donor red blood cells. ○ Passively acquired anti-A or anti-B
• Positive Antibody Screen • Compatible Crossmatches	○ Autoanti-IH (autoanti-H) or anti-Le^{bH} and non-group O units are selected. ○ Antibodies dependent on reagent red blood cell diluent. ○ Antibodies **demonstrating dosage** and donor red blood cells are from heterozygotes (i.e., expressing a single dose of antigen). ○ Donor unit is lacking corresponding antigen.

CAUSES OF POSITIVE PRETRANSFUSION TESTS

• Positive Antibody Screen • Incompatible Crossmatches, • Negative Autocontrol	○ Alloantibody(-ies)
• Positive Antibody Screen • Incompatible Crossmatches, • Positive Autocontrol, • Negative Direct Antiglobulin Test	○ Antibody to ingredient in enhancement media or enhancement-dependent autoantibody. ○ Rouleaux formation.
• Positive Antibody Screen • Incompatible Crossmatches, • Positive Autocontrol, • Positive DAT	○ Alloantibody causing either a delayed serologic or hemolytic transfusion reaction. ○ Passively acquired autoantibody (e.g., intravenous immune globulin). ○ Cold- or warm-reactive autoantibody. ○ Rouleaux formation

BLOOD PRESERVATION

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- The goal of blood preservation is to provide viable and functional blood components for patients requiring blood transfusion. RBC viability is a measure of in vivo RBC survival following transfusion.
- The FDA requires an average of 24-hour post-transfusion RBC survival of more than 75%

Braxton Hicks (1869)	Recommended sodium phosphate as blood preservative
Hustin (1914)	Use of sodium citrate as an anticoagulant solution for transfusion
Lewisohn (1915)	Determine the minimum amount of citrate needed for anticoagulation and demonstrated its non toxicity in small amounts
Rous and Turner (1930)	Introduced the citrate-dextrose solutions for preservation of blood
Loutit & Mollison (1943)	Introduced the formula for the preservative acid-citrate-dextrose (ACD)
Gibson (1957)	Introduced an improve preservative solution, Citrate-Phosphate-Dextrose (CPD), which was less acidic and eventually replaced ACD as the standard preservative used for blood storage

CHEMICALS IN ANTICOAGULANT SOLUTIONS (HARMENING)

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CHEMICAL	FUNCTION	PRESENT IN			
		ACD-A	CPD	CP2D	CPDA-1
Citrate (sodium citrate/citric acid)	Chelates calcium; prevents clotting.	X	X	X	X
Monobasic sodium phosphate	Maintains pH during storage; necessary for maintenance of adequate levels of 2,3-DPG.	X	X	X	X
Dextrose	Substrate for ATP production (cellular energy).	X	X	X	X
Adenine	Production of ATP (extends shelf-life from 21 to 35 days).				X

CHEMICALS IN ANTICOAGULANT SOLUTIONS (HENRY)

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CHEMICAL	FUNCTION
Citrate	Blood anticoagulant Binds calcium and prevents coagulation
Inorganic phosphate Buffer	Increase ATP production and RBC viability
Dextrose	Provide energy source for the RBC
Adenine	Improve the survival of stored donor RBC's Increased maximum shelf-life of blood from 21 days to 35 days; provides a SUBSTRATE for RBC to maximize ATP synthesis and thereby increase RBC viability
Citric Acid	Decreased pH, prevent caramelization

APPROVED BLOOD PRESERVATIVES

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APPROVE PRESERVATIVES	SHELF LIFE
Acid-Citrate-Dextrose (ACD)	21 Days
Citrate-Phosphate-Dextrose (CPD)	21 Days
Citrate-Phosphate-Double Dextrose (CP2D)	21 Days
Citrate Phosphate-Adenine (CPDA-1)	35 Days
Citrate-Phosphate-Dextrose-Adenine (CPDA-2)	42 Days

Note: The most commonly used anticoagulant-preservative solution for RBC is (CPDA-1)

RBC STORAGE LESIONS

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Table 1–2 RBC Storage Lesion

Characteristic	Change Observed
Viable cells (%)	Decreased
Glucose	Decreased
ATP	Decreased
Lactic acid	Increased
pH	Decreased
2,3-DPG	Decreased
Oxygen dissociation curve	Shift to the left (increase in hemoglobin and oxygen affinity; less oxygen delivered to tissues)
Plasma K^+	Increased
Plasma hemoglobin	Increased

Note: The loss of EBC viability has been correlated with the storage lesion, which is associated with various biochemical changes

ADDITIVE SOLUTIONS

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- Extends the shelf-life of RBC's to 42 days by adding nutrients
- Allows for the harvesting of more plasma and platelets from the unit
- Produces a packed RBC of lower viscosity that is easier to infuse

NAME	STORAGE TIME
Adsol (AS-1)	42 Days
Nutricel (AS-3)	42 Days
Optisol (AS-5)	42 Days
Solx (AS-7)	42 Days
COMPONENTS	
Saline	Where solutes are suspended
Adenine	Improve RBC survival
Glucose	ATP production
Mannitol	Acts as a RBC membrane stabilizer

ADDITIVE SOLUTIONS

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RBC Additives: Biochemical Characteristics

	AS-1	AS-3	AS-5	AS-7
Storage period (days)	42	42	42	42
pH (measured at 37° C)	6.6	6.5	6.5	6.6
24-hour survival (%)	83	85.1	80	80
ATP (% initial)	68	67	68.5	91
2,3-DPG (% initial)	6	6	5	1.5 µmol/L/g Hb
Hemolysis (%)	0.5	0.7	0.6	0.3

- RBC's stored for 42 days in AS-1, AS-3, on AS-5 demonstrated a mean of 24-hour post infusion survival of greater than 75%, the minimum requirement for satisfactory RBC survival

RBC REJUVENATION (for outdated blood)

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- Used in some blood centers to **REGENERATE ATP** and 2, 3 DPG
- Red cells stored in the liquid state for **FEWER THAN 3 DAYS AFTER THEIR OUTDATE** are rejuvenated for 1 to 4 hours at 37'C with the solutions
- Solutions contains PIGPA and/or PIPA:
  - > **PIGPA** (Phosphate, Inosine, Glucose, Pyruvate, Adenine)
  - > **PIPA** ((Phosphate, Inosine, Pyruvate, Adenine)
- **REJUVESOL** - only FDA approved rejuvenation solution

Note: Approx. 200 to 250 mg of IRON are contained in 1 RBC unit

RED CELL FREEZING

- For autologous units and storage of rare blood types using glycerol
- Frozen RBC shelf-life: 10 years

ADVANTAGES	DISADVANTAGES
Long-term storage (10 years)	A time-consuming process
Maintenance of RBC viability and function	Higher cost of equipment and materials
Low residual leukocytes and platelets	Storage requirements (-65'C)
Removal of significant amount of plasma proteins	Higher cost of products

RED CELL FREEZING

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ADVANTAGES	HIGH GLYCEROL	LOW GLYCEROL
1. Initial freezing temperature	-80'C	-196'C
2. Need to control freezing rate	No	Yes
3. Type of freezer	Mechanical	Liquid nitrogen
4. Maximum storage temperature	-65'C	-120'C
5. Shipping requirements	Dry Ice	Liquid nitrogen
6. Effect of changes in storage temperature	Can be thawed and refrozen	Critical

DEGLYCEROLIZATION

- Removal of glycerol after thawing, required to return the cells to a normal osmolality
- Blood unit is exposed to decreasing osmolarity, from hypertonic solution then followed by isotonic solution
- Thawing takes approx 45 minutes and involves immersing units into a 30'C to 37'C water bath and washing the RBC's with solutions of decreasing osmolarity
- Deglycerolized blood product should ensure **80% RBC MASS**

BLOOD SUBSTITUTES

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- Common feature: ability to carry)2 in the absence of intact red cells
- HEMOGLOBIN-BASED OXYGEN CARRIER
  - ➢ Chemically modified hemoglobin series
  - ➢ Recombinant hemoglobin
  - ➢ Encapsulated hemoglobin
  - ➢ Stroma-free hemoglobin series (SFHS)
- PERFLUOROCHEMICALS
  - ➢ Hydrocarbon structure which are H+ atoms have been replaced with fluorine
  - ➢ Immiscible with blood
  - ➢ Injected as emulsions with albumin

Note: Since RBC substitutes are drugs, they must go through extensive testing in order to obtain FDA approval. Safety and efficacy must be demonstrated through clinical trials.

PHASE OF TESTING

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PHASE	DESCRIPTION OF TESTING
Preclinical	In vivo and animal testing.
Phase I	Researchers test drug in a small group of people (20 to 80) for the first time to evaluate its safety, determine a safe dosage range, and identify side effects.
Phase II	The drug is given to a larger group of people (100 to 300) to see if it is effective and to further evaluate its safety.
Phase III	The drug is given to large groups of people (1,000 to 3,000) to confirm its effectiveness, monitor side effects, compare it to commonly used treatments, and collect information that will allow the drug to be used safely.
Phase IV	Postmarketing studies to gather additional information about the drug's risks, benefits, and optimal use.

ADVANTAGES	DISADVANTAGES
Long shelf-life	Short intravascular half-life
Very stable	Possible toxicity
No antigenicity (unless bovine)	Increased O2 affinity
No requirement for blood typing procedures	Increased oncotic effect